

Liam C Pendleton, MSc

Quantitative Biologist

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Personal Objective

To contribute to innovative, data-informed conservation solutions by combining a background in quantitative ecology and community-driven science. I aim to turn complex ecological data into practical tools and strategies that address conservation challenges and empower community involvement in protection and restoration of the environment.

Skills

Data Analytics | Statistical Modeling | Machine Learning | Data Visualization | Spatial Analysis | GIS | R | SQL | GitHub
Excel | ESRI | GIS | Project Management | Field Work | Community Outreach | Habitat Surveys | Plant ID | Soil Science
Regulatory Compliance | Environmental Monitoring | Cross-Sector Collaboration | Technical Reporting | Team Supervision

Professional Experience

Washington Cooperative Fish and Wildlife Research Unit

Seattle, WA

Research Assistant

07/2021 - 06/2024

Collaborated with state, federal and community science partners to design and conduct a long-term seabird monitoring program at Protection Island National Wildlife Refuge, WA. Used Bayesian modeling, GIS, and machine learning techniques to describe reproductive parameters and behavior of two migratory nesting seabird species.

- Managed logistics for field operations including equipment procurement, scheduling, and resource allocation
- Organized and trained field crew; directed 20+ field days per season at a remote island in adherence to ESA protocols while in proximity to ESA-listed species
- Authored technical reports, permit applications, data collection and safety protocols, and funding applications
- Built hierarchical models in a Bayesian framework using R and JAGS to describe seabird reproductive success
- Used hidden Markov modeling in R to identify state-specific behaviors and foraging habitat of nesting seabirds
- Organized and participated in a various scientific communication events targeting audiences of varying levels of expertise

Bird Laboratory, Eastern Michigan University

Ypsilanti, MI

Research Assistant

08/2019 - 05/2021

Conducted a spatial analysis of habitat use by a protected songbird species, using R-based modeling and spatial data to define and identify core habitat. Highlighted potential conservation priorities across 5,000+ hectares of habitat.

- Managed a large-scale spatial dataset describing habitat usage across 5,000+ hectares; validated and standardized data
- Used kernel density estimations and Brownian bridge movement models in R to describe habitat and behavior
- Analyzed .wav files using Audacity to describe energy expenditure of breeding and non-breeding passerines
- Led logistical planning for a multi-week field data collection trip along the US west coast, including travel coordination, equipment procurement, and scheduling for team members
- Collected morphometric and physiological data from 12 birds, adding ~20% more data points to existing datasets

Aquatic Geochemistry Laboratory, University of Michigan

Ann Arbor, MI

Data Analyst

02/2020 - 09/2020

Identified primary drivers of methane-producing reactions in permafrost samples from the Alaskan Arctic.

- Used R and Excel to assess annual variation in 15+ chemical properties of 50+ permafrost samples, identifying sunlight, bacterial activity and iron concentration as significant contributors to methane production
- Provided weekly data summaries to the principal investigator, resulting in improved efficiency of ongoing lab projects and project workflow
- Used GitHub to track and manage branches of analytical code

Bird Center of Washtenaw County
Supervisor

Ann Arbor, MI
05/2017- 11/2019

Oversaw patient data management and tracking, leading database optimization, digital system implementation, and data-driven decision-making to improve operational efficiency and patient care.

- Hired, supervised and trained 20+ interns and volunteers; created training materials and provided professional development
- Coordinated annual fundraising and community outreach events, securing \$30,000+ in donations annually
- Developed event and operational budgets; forecasted expenses, tracked costs, and oversaw equipment procurement
- Collaborated with board members and external partners to build partnerships supporting the Bird Center's growth
- Managed a database containing 3,000+ patient records, overseeing data entry and retrieval processes
- Wrote annual reports describing clinic statistics including seasonal trends in patient circumstances, rehab success rates, and species treated

Detroit Audubon Society
Scientific Technician

Detroit, MI
05/2018 - 08/2019

- Monitored 30+ shorebird nests across multiple sites, tracking and reporting nest and environmental conditions
- Conducted habitat and vegetation surveys and recorded water quality at study sites
- Assisted in restoring 3 acres of native prairie habitat, increasing habitat quality for native bird and pollinator species
- Engaged with community members to promote local conservation methods and awareness of native wildlife

Education

University of Washington | 2024
M.S., Aquatic and Fishery Sciences

Eastern Michigan University | 2021
B.S., Biology

Washtenaw Community College | 2018
A.A., Liberal Arts

Volunteering

Pacific Seabird Group
Organizer

Seattle, WA
11/2023 - 03/2024

- Designed and ran a professional development panel targeted at early career scientists. Contacted and organized panelists from varying professional backgrounds, wrote starter questions, and facilitated dialogue between panelists and audience.
- Assisted in organizing a professional mentoring event. Contacted and confirmed mentors, and paired with student mentees.

Salish Sea Guillemot Network
Field Technician

Seattle, WA
05/2022 - 08/2023

- Monitored foraging activity of pigeon guillemots at various sites around Puget Sound, WA. Identified prey species, nest locations, and provisioning frequency.

Ann Arbor Natural Area Preservation
Field Technician

Ann Arbor, MI
05/2019 - 08/2020

- Conducted point counts and vegetation surveys at 10+ sites around Ann Arbor, MI to assess songbird abundance and diversity following habitat restoration initiatives.